Tingxi Li

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## Research Interests

I am broadly interested in the **inference-time efficiency** of deep learning systems across two levels: (i) kernel and subgraph-level efficiency, and (ii) model-level dynamicity. My primary research interests lie in **deep learning compilation**, targeting compiler infrastructure and AI-driven kernel generation. I also investigate dynamic behavior vulnerabilities in deep learning models and develop defenses against efficiency exploitation.

## Publications

**Characterizing Real-World Bugs in Tile Programs for Automated Bug Detection**
*R. Rathnasuriya, Z. Song, N. Majoju, **T. Li**, A. Moharir, W. Yang, T. Xie* — *ISSTA'26*

**SoK: Efficiency Robustness of Dynamic Deep Learning Systems**
*R. Rathnasuriya, **T. Li**, Z. Xu, Z. Song, M. Haque, S. Chen, W. Yang* — *USENIX Sec'25*

**COMET: Closed-loop Orchestration for Malicious Elicitation Techniques in Code Models**
*Z. Xu, **T. Li**, R. Rathnasuriya, Z. Song, J. Ren, B. Mandalapu, S. Setayeshpour, X. Du, W. Yang* — *Technical report*

**Identify then Exploit: Degrading Performance of Vision-based Deep Learning Systems**
***T. Li**, M. Ji, R. Rathnasuriya, S. Chen, Y. Hu, W. Yang* — *Under review*

## Research Projects

### A Survey of Automation in Kernel Generalization and Optimization — Jan. 2026 – Present

- Systematized DL compilation research into AI-driven kernel generation and compiler infrastructure, with subcategories on search space pruning, auto-tuning, cost modeling, and candidate validation.
- Analyzed IR evolution from loop-based to tile-based abstractions and its downstream impact on the compilation ecosystem.

### TileBench: A Comprehensive Benchmark for Tile-Based DSLs — Mar. 2026 – Present

- Built a benchmark covering PyTorch Eager/Compile and Triton/TileLang across diverse hardware and workloads, filling gaps left by existing AI-kernel suites.
- Merged KernelBench and TritonBench into a unified dataset with verified correctness and a multi-dimensional kernel taxonomy (functionality, complexity, etc.).

### Amazon Trusted AI Competition — Nov. 2024 – Jul. 2025

- Finalist team ($250K prize) targeting black-box jailbreaking of CodeLLMs.
- Fine-tuned surrogate models for adversarial prompt guidance, achieving 0.80+ recall and 0.75+ F1 in malicious intent detection.

## Education

**University of Texas at Dallas** — **Sep. 2024 – Present**
*Doctor of Philosophy in Computer Science*

**Dalian University of Technology** — **Sep. 2019 – May 2024**
*Bachelor of Science*

## Internship Experience

**Sophgo** — **Jun. 2024 – Aug. 2024**
*Research Intern* — *Shenzhen, China*

- Developed C++ API code for models on RISC-V processors and authored deployment documentation.
- Fine-tuned models on private datasets, identified failure cases, and applied data augmentation to improve robustness.

## Miscellaneous

**Tech Stack**: Python; C; C++; Java; PyTorch; LaTeX; SQL